Spring-constant scales and units in elastic network models

Notes for penm

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1 The question

In a harmonic network with $k_{ij} = \gamma\,\kappa_{ij}$, where γ is an overall scale and κ_{ij} the model's shape, temperature and stiffness enter only through $\beta\gamma$, in Å^{-2}. Mode shapes, structural responses $\mathbf{K}^{+}\mathbf{f}$ and differences of βTS between two networks are independent of both. Fluctuations scale as $(\beta\gamma)^{-1}$, βV as $\beta\gamma$, and absolute entropic terms $\ln(\beta\lambda/2\pi)$ depend on $\beta\gamma$ and on the length unit. A free energy that adds an energy to an entropic term therefore needs γ in physical units, consistent with the units of β.

These notes record what five papers say about the scale γ, its units, what it was fitted to, and at what temperature.

2 The papers

Ming & Wall 2005 [1]

Fits C_α networks of trypsinogen to a CHARMM22 all-atom harmonic model (vacuum minimisation, Hessian computed in CHARMM). Two targets: the Kullback–Leibler divergence D between the C_α distributions, and the C_α mean-square displacements (MSDs). The optimal γ for a given network follows in closed form from D (their eq. 10).

model	target	γ (kcal/mol/Å^2)	r_c (Å)	ϵ
ANM	D	53.4	7.75	—
ANM	MSDs	0.4	15.4	—
BENM	D	4.26	10.5	42
BENM	MSDs	1.2	10.5	42

BENM multiplies the spring constant of backbone neighbours by ϵ. The reference is harmonic, so its fluctuations scale with k_BT exactly as the network's do: these γ are mechanical stiffnesses, independent of temperature (the MSDs are quoted at 300 K). The D-optimal ANM "exhibits poor MSDs"; fitting the MSDs instead moves γ by a factor of about 130 and the cutoff from 7.75 to 15.4 Å. One protein.

Orellana et al. 2010 [2]

ED-ENM, fitted to essential dynamics of MD metatrajectories (AMBER, CHARMM, OPLS/AA; ≥ 10 ns; six training proteins, 32+ test proteins). Sequence neighbours up to $i{+}3$ get $K_{ij} = C_{\text{seq}}/S_{ij}^2$; other pairs within a size-dependent cutoff (about 8–16 Å) get an inverse sixth power of distance. The

constants are $C_{\mathrm{seq}} = 60$ and $C_{\mathrm{cart}} = 6$ kcal/(mol Å^2), found by matching total variance, variance profile and reduced variance of the MD essential modes, and robust to ±10 and ±1 respectively. The long-range term is written $C_{\mathrm{cart}}/d_{ij}^6$ in eq. 6 and $(C_{\mathrm{cart}}/d_{ij})^6$ in eq. 8; only the first is consistent with the stated units. The energy is $E = \sum K_{ij}(R_{ij} - R_{ij}^0)^2$, without the factor $\frac{1}{2}$. The MD temperature is not stated in the main text. Calibrating against MD gives effective constants at the MD temperature, through $K_{ij}^{\mathrm{app}} = k_B T/\langle (R_{ij} - R_{ij}^0)^2 \rangle$ and $K_\nu = k_B T/\lambda$.

Leioatts et al. 2012 [3]

Fits several spring functions (cutoff, exponential, HCA-like, bonded/nonbonded) to > 1 μs MD of three GPCRs. The inverted ENM eigenvalues "are normalized … such that the total amplitude of predicted motion is identical" to MD, so the fits determine the shape of $k(r)$, not its scale; the cutoff model uses $k = 1$. The potential is $U_{ij} = k(R_{ij}^\circ)(R_{ij} - R_{ij}^\circ)^2$, again without $\frac{1}{2}$. Main result: after fitting, all spring functions perform alike, because all make the first shell of neighbours (about 3.8 Å) much stiffer than more distant contacts, by about an order of magnitude or more.

Fuglebakk et al. 2013 [4]

Benchmarks ANM (cutoffs 0.8 and 1.8 nm), GNM, HCA, pfANM, REACH and βGM against covariances from MD for seven proteins (five run at 300 K; two taken from earlier publications). Every covariance is normalised by its trace, which "removes a global amplitude factor from the fluctuations and covariances, which none of the ENMs describes realistically anyway"; of γ: "its value is not important". No scale is determined. Results bearing on parameters:

- the ANM cutoff that best reproduces MD covariances lies in 0.8–1.0 nm; cutoffs that best reproduce B-factors are much longer and suppress collective motion;
- models with weak long-range interactions (HCA, REACH, $\mathrm{ANM}_{0.8}$, βGM) beat those with strong ones ($\mathrm{ANM}_{1.8}$, pfANM); the precise decay law matters little;
- B-factors are discouraged as a benchmark, and "are not necessarily proportional to the temperature".

REACH is used with parameters from MD of a myoglobin dimer.

Dubanevics & McLeish 2022 [5]

ANM and BENM of three proteins (CAP, GST, Mpro) compared with 300 K MD (PCA of mass-weighted covariance). Potential $V_{ij} = \frac{k}{2}(r_{ij} - r_{ij}^{(0)})^2$ within r_c, starting from $k = 1$ kcal/(mol Å^2), then rescaled so that the first ENM mode frequency matches the first MD mode (their eq. 3, as printed):

$$k_{\mathrm{scaled}} = k\,\frac{\omega_{\mathrm{MD}}(1)}{\omega_{\mathrm{ENM}}(1)}.$$

The preceding sentence states $\omega^2 \propto k$, so matching the frequencies requires the square of this ratio; whether the tabulated values used the printed or the squared form cannot be told from the paper. Frequencies come from the mass-weighted Hessian and enter a quantum harmonic free energy, so the absolute scale matters there. MD: 300 K, 300 ns production, PCA on the last 50–150 ns (their Table 5).

protein	r_c (Å)	ANM ($\epsilon = 1$)	BENM ($\epsilon = 100$)
CAP	8.0	31.10	23.56
GST	8.5	25.71	20.55
Mpro	8.5	12.28	9.01

(k_{scaled} in kcal/mol/Å^2, their Table 3.) The optimal ϵ is protein-dependent (100 for CAP and GST, 200 for Mpro), and the low:high frequency ratio across the BENM gap is 2:1, as in Ming & Wall.

3 Observations

1. **Most comparisons are scale-free by construction.** Leioatts and Fuglebakk normalise amplitudes before comparing; they constrain the shape of $k(r)$ and the cutoff, not γ.
2. **Three papers give an absolute scale, by different routes.** Against an all-atom *harmonic* model (Ming & Wall): mechanical stiffness, independent of temperature. Against *MD* variances or frequencies (Orellana; Dubanevics & McLeish): effective constants at the MD temperature, absorbing anharmonicity and finite sampling.
3. **For a uniform ANM, γ is not a single number.** It depends on the target and is coupled to the cutoff: 53.4 at 7.75 Å (distribution shape), 12–31 at 8–8.5 Å (first MD mode), 0.4 at 15.4 Å (MSDs), all in kcal/(mol Å^2). Across proteins at fixed protocol it varies by a factor of about 2.5 (Dubanevics & McLeish). The conventional $\gamma = 1$ kcal/(mol Å^2) is none of these.
4. **Conventions differ by factors of 2.** $\frac{1}{2}k\,\Delta d^2$ (Dubanevics & McLeish; penm) versus $k\,\Delta d^2$ (Leioatts; Orellana). A constant taken from one paper into another code must be converted.
5. **B-factors are a poor calibration target.** Ming & Wall, Orellana and Fuglebakk all note that crystallographic B-factors include static disorder, lattice effects and crystal constraints; Fuglebakk finds that fitting to them selects networks with little collective motion.
6. **Shape is better settled than scale.** Leioatts and Fuglebakk agree that what matters is a strong contrast between the first neighbour shell (or sequence neighbours) and weaker, fast-decaying long-range springs; the functional form matters little.

4 penm's models

As of penm 0.6.0, every model returns spring constants in kcal/(mol Å^2), so all energies are in kcal/mol and `beta_boltzmann()` (mol/kcal, CODATA gas constant) applies to every model. Models published in kJ keep their published numbers in the code and are converted on return.

`model`	published units of k	source; in penm
`ming_wall`	kcal/(mol Å^2)	[1]; default $k = 4.26$ (was 4.5 before 0.6.0)
`reach`	kJ/(mol Å^2), MD at 300 K	Moritsugu et al. 2009, Table 1, solvated dimer; converted
`hnm`	kJ/(mol Å^2)	Hinsen et al. 2000, as in bio3d `ff.calpha`; converted; C_α nodes only
`anm`, `gnm`, `pfanm`, `hnm0`	none	read as kcal/(mol Å^2), arbitrary scale ($k = 1$, $d_{\max} = 10$ Å for `anm`)

For the arbitrary-scale models, energies and anything that adds an energy to an entropic term carry that arbitrary scale; the calibrations of Section 2 are the alternatives.

References

[1] D. Ming and M. E. Wall, Allostery in a coarse-grained model of protein dynamics, *Phys. Rev. Lett.* **95**, 198103 (2005).

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[3] N. Leioatts, T. D. Romo and A. Grossfield, Elastic network models are robust to variations in formalism, *J. Chem. Theory Comput.* **8**, 2424 (2012).

[4] E. Fuglebakk, N. Reuter and K. Hinsen, Evaluation of protein elastic network models based on an analysis of collective motions, *J. Chem. Theory Comput.* **9**, 5618 (2013).

[5] I. Dubanevics and T. C. B. McLeish, Optimising elastic network models for protein dynamics and allostery: spatial and modal cut-offs and backbone stiffness, *J. Mol. Biol.* **434**, 167696 (2022).